

# Luis Mastrangelo

Senior Software Engineer

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## Professional Summary

I am Senior Software Engineer with professional and academic experience in Switzerland, Argentina, France and USA. I have worked for more than 15 years in the private industry in different roles, as Software Engineer, Network Administrator and Freelance Developer. I delivered many software projects for a wide variety of clients across different sectors.

I have a Ph.D. in Informatics at the University of Lugano, Switzerland, where I was able to perform outstanding research in programming languages, runtime systems and virtual machines.

In the last years I have been working in the Web3 space. I worked with Ethereum, NEAR, Avalanche, EOS/WAX and Everscale (TON) using mainly Solidity, Rust and TypeScript/JavaScript.

## Education

- Apr 2013–June 2019 **Ph.D. Graduate, Computer Science, University of Lugano, Switzerland**  
Programming languages research in Java, Scala and Haskell, runtime systems and virtual machines with C++, software verification, data analysis and visualization using Python and R.
- Apr 2007–Nov 2011 **Licentiate in Computer Science, University of Buenos Aires, Argentina**  
This degree is considered by many institutions as being equivalent to an M.Sc. degree.
- Apr 2011–Oct 2011 **Internship, INRIA/University of Strasbourg, France**  
Internship title: "A Virtual Machine for Automatic Program Parallelization".
- Apr 2003–Dec 2006 **Computer Analyst, University of Buenos Aires, Argentina**  
This is a 3 years degree as part of the *Licentiate in Computer Science* programme.

## Work Experience

- Nov 2023–Mar 2026 **Senior Software Engineer, Hashgraph (ex. Swirlds Labs), Remote**  
Part of the Smart Contracts team, building tools to support EVM development in Hedera networks. While working here I
- created the [Hedera Forking](#) project, which allows developers to fork test their contracts against Hedera System Contracts,
  - contributed to setup a custom Sourcify instance, which allowed developers to verify their Smart Contracts,
  - refactored Hashgraph's JSON-RPC Relay backend so it can be more easily extensible,
  - integrated the Ethereum Execution Client testing framework in Python/Pytest into Hedera's own EVM,
  - debugged production issues while on-call using Grafana and GCP, and
  - added support to onboard new clients to Hashgraph's TheGraph custom instance.

- Apr 2023–May 2023 **Senior Software Engineer, ChainSafe**, Remote  
I worked in Lodestar, an Ethereum Consensus Client written in TypeScript. I helped to fix some small bugs while working here.
- Jul 2022–Jan 2023 **Senior Blockchain Engineer, ConsenSys**, Remote  
Part of the Truffle team, a development environment to build and test Solidity Smart Contracts. I focused on their VS Code extension for Truffle. While working here I  
  - added and tested new features to the extension using TypeScript, and
  - helped with the release workflow in GitHub Actions.
- May 2022–Jun 2022 **Lead Blockchain Engineer, Campground**, Remote  
Smart Contract development with Rust and TypeScript for the NEAR blockchain. While working here I  
  - built a system in Rust to automatically generate public docs for our smart contracts,
  - tested and deployed the main smart contracts, and
  - helped with the integration of the Smart Contracts into our backend using TypeScript.
- Sep 2020–Apr 2022 **Senior Software Engineer, EPAM Systems**, Zurich, Switzerland  
Ethereum & NEAR blockchains Smart Contract development with Solidity, Rust and TypeScript. Also architected a greenfield marketplace application.
- Aug 2019–Jan 2020 **Technical Consultant, DB&L Services SA for Avaloq**, Lugano, Switzerland  
IT infrastructure support for Monitoring Operations at Avaloq Sourcing (CH & LI) SA.
- Dec 2006–Mar 2011 **Project Leader, Recursiva S.R.L.**, Buenos Aires, Argentina  
Software engineering and architect roles using Java and C# among others technologies. Besides building and testing tasks, I have visited customers and potential customers to gather requirements to implement them in our systems.
- Mar–Jul 2006 **Network Administrator, Texpampa and HS Eventos**, Buenos Aires, Argentina  
Technical service, software installation and maintenance of company network.
- Mar 2000–May 2004 **Software Developer, Fernando Hermo y Asoc.**, Buenos Aires, Argentina  
ERP and CRM development using VB 6, SQL Server/Microsoft Access and Crystal Reports. Technical support to customers both on-line and on-site.

## Freelance & Open Source Projects

- Apr 2023–Mar 2025 **Software Engineer, SEVM Open Source project**, Remote  
A Symbolic EVM bytecode interpreter, parser and decompiler written in TypeScript. This work involves parsing and analyzing bytecode to build its corresponding CFG, to then extract method information or decompile to either Solidity or Yul.
- Jan–May 2020 **Software Engineer, GTM (cont.) and GTM Dashboard Open Source**, Remote  
Added support for submodules to Git Time Metric (Golang) [PR](#) and built a dashboard for GTM from scratch (Rust, TypeScript and Svelte).
- Feb 2014–Jan 2020 **Software Developer, JNIF**, Lugano, Switzerland  
Java Native Instrumentation Framework written in C++. JNIF allows to analyze and instrument Java bytecode, including JDK objects, something it is not possible with other instrumentation frameworks. It parses the bytecode and reconstruct its CFG so it can be instrumented in later phases.
- May–Sep 2018 **Software Developer, Essentia, Family Office**, Lugano, Switzerland  
Desktop application written in C# and Scala.
- Apr 2016–Oct 2017 **Software Developer, HSC**, Lugano, Switzerland  
A Supercompiler (similar to symbolic execution) prototype for a Haskell-like language.

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## Technical Skills

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|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming Languages | Rust, JavaScript/TypeScript, Python, C/C++, Solidity/Yul, Java, Kotlin, Scala, Haskell, Golang, R, C#, Assembler, Prolog, SQL, PHP, HTML/CSS, UML |
| Paradigms & Fields    | OOP, Functional Programming, Static & Dynamic Analysis, Data Science                                                                              |
| Databases             | PostgreSQL, MySQL, Oracle, Datalog, SQL Server, SQLite, Redis, Design                                                                             |
| Editors               | VSCode, IntelliJ/CLion/Rider, Eclipse, Visual Studio, Xcode, Vim/Emacs                                                                            |
| VCSs                  | Git, GitHub, GitLab, BitBucket, Mercurial, Subversion, TortoiseSVN, CVS                                                                           |
| DevOps                | Docker, Podman, Kubernetes, Helm, Lima, GitHub Actions, GitLab CI/CD                                                                              |
| Tools                 | GitHub Copilot, Node.js, LLVM, Apache/Nginx, Azure, GCP, REST, GraphQL                                                                            |
| Project Management    | Scrum, GitHub Projects, Agile, JIRA, Kanban Boards                                                                                                |
| Blockchain & Web3     | Ethereum, NEAR, Avalanche, Smart Contracts, Foundry, Hardhat, TheGraph                                                                            |

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## Language Proficiency

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| English | <b>Proficient User:</b> Level C2 under <a href="#">CEFR</a>  |
| Spanish | <b>Native speaker</b>                                        |
| Italian | <b>Proficient User:</b> Level C1 under <a href="#">CEFR</a>  |
| French  | <b>Independent User:</b> Level B1 under <a href="#">CEFR</a> |

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## Teaching Experience

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|-------------------|----------------------------------------------------------------------------------------------|
| Apr 2013–Nov 2017 | <b>Teaching Assistant</b> , <i>University of Lugano</i> , Switzerland. Multiple courses.     |
| Apr 2006–Nov 2010 | <b>Teaching Assistant</b> , <i>University of Buenos Aires</i> , Argentina. Multiple courses. |

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## Publications DBLP

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|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| University of Lugano           | <b>Luis Mastrangelo</b> , Ph.D. Thesis, Advisors: Prof. M. Hauswirth and Prof. N. Nystrom. <a href="#">When and How Java Developers Give Up Static Type Safety</a> . 2019     |
| University of Lugano           | <b>Luis Mastrangelo</b> , M. Hauswirth and N. Nystrom. <a href="#">Casting about in the Dark: A Survey of Cast Usage in Java</a> . 2019                                       |
| University of Lugano           | <b>Luis Mastrangelo</b> , L. Ponzanelli, A. Mocci, M. Lanza, M. Hauswirth and N. Nystrom. <a href="#">Use at Your Own Risk: The Java Unsafe API in the Wild</a> . 2015        |
| University of Lugano           | <b>Luis Mastrangelo</b> and M. Hauswirth. <a href="#">JNIF: Java Native Instrumentation Framework</a> . 2014                                                                  |
| INRIA/University of Strasbourg | A. Jimborean, <b>Luis Mastrangelo</b> , V. Loechner and P. Clauss. <a href="#">VMAD: an Advanced Dynamic Program Analysis &amp; Instrumentation Framework</a> . 2012          |
| INRIA/University of Strasbourg | A. Jimborean, P. Clauss, B. Pradelle, <b>Luis Mastrangelo</b> and V. Loechner. <a href="#">Adapting the Polyhedral Model for Efficient Speculative Parallelization</a> . 2012 |